Durham University

EXAMINATION PAPER

Examination Session:	Year:	Exam Code:
May/June	2016	BIOL1171-WE01

Title:

Genetics

Time Allowed:	2 hours	
Additional Material provided:	None	
Materials Permitted:	None	
Calculators Permitted:	Yes	Models Permitted: Casio fx-83 GTPLUS or Casio fx-85 GTPLUS calculator.
Visiting Students may use dictionaries:	No	

Instructions to Candidates:	Answer **all** questions in **Section A** (all 8 questions carry equal marks with a total of 33% of the overall marks). Answer **two** questions from **Section B** (67% of the marks). Answer all questions in Section A in the **space provided** and each question from section B in a **separate** answer book. You are advised to spend **40 minutes** on Section A and **40 minutes** on **each** question in Section B**.** Use chemical equations and/or labelled diagrams, where appropriate, to illustrate your answers.

Revision:	

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Section A: Answer all questions in this section

1) Draw, in appropriate detail, the structure of the DNA double helix.

2) Describe, briefly, the chain of reactions that make up the Polymerase Chain Reaction (= PCR).

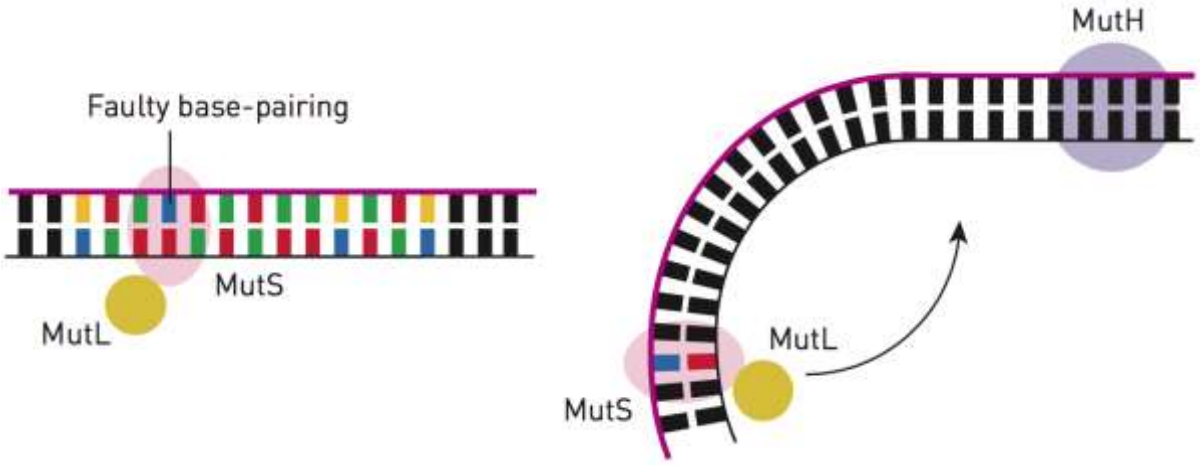
Answer:

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3) With the aid of the simplified Figure 1, explain how the MutHLS system repairs mismatches correctly.

Figure 1.



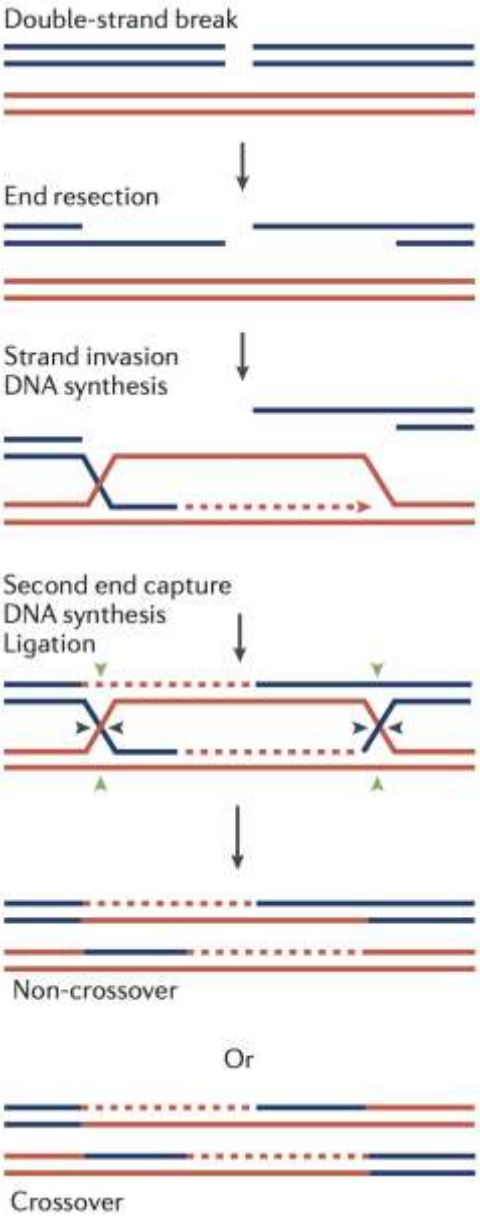
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4) With the help of the Figure 2, explain how the bacterial enzymes RecA, RecBCD and RuvABC initiate and complete double-stranded DNA break repair to generate crossover and non-crossover recombinant products.

Figure 2.



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5) Bacterial ribosomes consist of 30 and 50S subunits. What does the S stand for, and what does it mean?

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6) What is a gene?

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7) What are sigma factors? What do they do?

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8) A couple who are about to give birth learn that, in *both* their families, both of their unaffected parents have siblings with cystic fibrosis (a rare autosomal recessive disease). No other family members are known to have been affected, including the couple themselves. Showing your working, calculate the probability that the couple's child will have cystic fibrosis.

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Section B: Answer two questions in this section

1) Describe how *either* the lac operon *or* the trp operon is regulated.

2) Explain how translation is regulated in bacteria.

3) How can heritable changes in ploidy arise? What effects can these have on the offspring to whom they are passed?

4) What can genetics tell us about *either* the study of human evolution *or* the study of human disease?

End

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